

Embedding and quotient results for the natural maps $E_{m,n}$ between Rochberg families

FA/Banach run packet for arXiv:1906.06677

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Source problem

The source paper is F. Cabello Sanchez, J. M. F. Castillo, and W. H. G. Correa, *Higher order derivatives of analytic families of Banach spaces*, arXiv:1906.06677.

In Section 7, after defining the natural maps

$$E_{m,n} : \mathcal{F}_z^{(m+n-1)} \longrightarrow F[m, n],$$

the authors prove that $E_{m,2}$ and $E_{2,m}$ are embeddings with complemented range. They then write that it is tempting to conjecture the same complemented-range picture for every $m, n \geq 2$, and add:

“We do not even know whether $E_{3,3}$ is an embedding or if $F[3, 3]$ has a subspace isomorphic to $\mathcal{F}_z^{(5)}$.”

This packet proves the embedding half for all m, n . It also identifies the quotient as $F[m-1, n-1]$ when $m, n \geq 2$. It does not prove complemented range.

Definitions and notation

Let $\mathcal{F}$ be an acceptable space of analytic functions, and fix a base point z . Write $\mathcal{F}_z^{(r)}$ for the r -th Rochberg space at z . The standard Rochberg exact sequences have the form

$$0 \longrightarrow \mathcal{F}_z^{(a)} \xrightarrow{\iota_{a,a+b}} \mathcal{F}_z^{(a+b)} \xrightarrow{\pi_{a+b,b}} \mathcal{F}_z^{(b)} \longrightarrow 0.$$

In particular, $\iota_{a,a+b}$ is an isomorphic embedding and $\pi_{a+b,b}$ is a quotient map.

Fix $m \geq 1$. Let $\mathcal{T}$ be the acceptable space supplied by the source paper so that

$$\mathcal{T}_z^{(1)} = \mathcal{F}_z^{(m)}.$$

Following the source notation, put

$$F[m, n] = \mathcal{T}_z^{(n)}.$$

The corresponding exact sequences are

$$0 \longrightarrow F[m, a] \xrightarrow{\iota_{a,a+b}^m} F[m, a+b] \xrightarrow{\pi_{a+b,b}^m} F[m, b] \longrightarrow 0.$$

The source defines $E_{m,n}$ by taking a jet

$$x = (x_{m+n-2}, \dots, x_0) \in \mathcal{F}_z^{(m+n-1)}$$

and sending it to the $n \times m$ matrix

$$E_{m,n}(x) = \left(\binom{i+j}{i} x_{i+j} \right)_{0 \leq i < n, 0 \leq j < m},$$

with the same ordering convention as in the paper. The source already observes that each $E_{m,n}$ is injective and continuous.

Idea of the proof

The map $E_{m,n}$ respects the two natural “drop the highest derivative” exact sequences. Dropping the highest row in $F[m, n]$ gives $F[m, n-1]$, while dropping the highest coordinate in $\mathcal{F}_z^{(m+n-1)}$ gives $\mathcal{F}_z^{(m+n-2)}$. The only new kernel coordinate maps into $F[m, 1] = \mathcal{F}_z^{(m)}$ as a nonzero scalar multiple of the standard inclusion $\mathcal{F}_z^{(1)} \hookrightarrow \mathcal{F}_z^{(m)}$.

Thus a simple Banach-space diagram lemma says: if the left and right vertical maps in a short-exact-sequence diagram are embeddings, then so is the middle one. Starting from $E_{m,1} = \text{id}_{\mathcal{F}_z^{(m)}}$, induction on n proves that every $E_{m,n}$ is bounded below.

For the quotient, the right invariant is a first-order defect map. If $C = (c_{m-1}, \dots, c_0)$ is a function in the order- m Rochberg family, then the expression

$$c'_j - (j+1)c_{j+1}$$

measures the failure of C to be the jet tuple $\tau_{(m,0]}f$ of one function $f \in \mathcal{F}$. Passing from n -jets of C to $(n-1)$ -jets of these defects gives a bounded quotient map $F[m, n] \rightarrow F[m-1, n-1]$. Its kernel is exactly $E_{m,n}[\mathcal{F}_z^{(m+n-1)}]$.

A diagram lemma

Lemma 1. *Consider a commutative diagram of Banach spaces with exact rows*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K & \xrightarrow{i} & Z & \xrightarrow{q} & X & \longrightarrow & 0 \\ & & \downarrow A & & \downarrow B & & \downarrow C & & \\ 0 & \longrightarrow & Y & \xrightarrow{j} & W & \xrightarrow{p} & V & \longrightarrow & 0. \end{array}$$

Assume A and C are isomorphic embeddings. Then B is an isomorphic embedding.

Proof. Continuity of B is part of the diagram. We prove a lower estimate. Let M_C and M_A be constants such that

$$\|x\|_X \leq M_C \|Cx\|_V, \quad \|k\|_K \leq M_A \|Ak\|_Y.$$

Let M_j be a lower-embedding constant for j , so

$$\|y\|_Y \leq M_j \|jy\|_W.$$

Fix $z \in Z$. Since q is a quotient map,

$$\text{dist}(z, iK) = \|qz\|_X.$$

The right square gives

$$\|qz\|_X \leq M_C \|Cqz\|_V = M_C \|pBz\|_V \leq M_C \|p\| \|Bz\|_W.$$

Choose $k \in K$ with

$$\|z - ik\|_Z \leq 2M_C \|p\| \|Bz\|_W.$$

Using commutativity on the left square,

$$jAk = Bik = Bz - B(z - ik).$$

Therefore

$$\begin{aligned} \|k\|_K &\leq M_A \|Ak\|_Y \leq M_A M_j \|jAk\|_W \\ &\leq M_A M_j (\|Bz\|_W + \|B\| \|z - ik\|_Z) \leq C_1 \|Bz\|_W \end{aligned}$$

for a constant C_1 depending only on the diagram. Hence

$$\|z\|_Z \leq \|z - ik\|_Z + \|i\| \|k\|_K \leq C_2 \|Bz\|_W.$$

This proves that B is bounded below, and therefore an isomorphic embedding. $\square$

Main theorem

Theorem 1. *For every acceptable space $\mathcal{F}$, base point z , and integers $m, n \geq 1$, the natural map*

$$E_{m,n} : \mathcal{F}_z^{(m+n-1)} \longrightarrow F[m, n]$$

is an isomorphic embedding.

Proof. Fix m . We induct on n . For $n = 1$, the space $F[m, 1]$ is $\mathcal{F}_z^{(m)}$, and $E_{m,1}$ is the identity up to the harmless convention identifying the single row with the m -tuple. Thus $E_{m,1}$ is an isomorphic embedding.

Assume that $E_{m,n-1}$ is an isomorphic embedding for some $n \geq 2$. Consider the exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}_z^{(1)} & \xrightarrow{\iota_{1,m+n-1}} & \mathcal{F}_z^{(m+n-1)} & \xrightarrow{\pi_{m+n-1,m+n-2}} & \mathcal{F}_z^{(m+n-2)} \longrightarrow 0 \\ & & \downarrow A_{m,n} & & \downarrow E_{m,n} & & \downarrow E_{m,n-1} \\ 0 & \longrightarrow & F[m, 1] & \xrightarrow{\iota_{1,n}^m} & F[m, n] & \xrightarrow{\pi_{n,n-1}^m} & F[m, n-1] \longrightarrow 0. \end{array}$$

Here

$$A_{m,n} = \begin{pmatrix} m+n-2 \\ m-1 \end{pmatrix} \iota_{1,m} : \mathcal{F}_z^{(1)} \longrightarrow F[m, 1] = \mathcal{F}_z^{(m)}.$$

The scalar is nonzero, and $\iota_{1,m}$ is a standard Rochberg embedding, so $A_{m,n}$ is an isomorphic embedding.

It remains only to check commutativity. The right square commutes because deleting the highest row of the matrix $E_{m,n}(x)$ is exactly the matrix $E_{m,n-1}$ applied to the jet obtained by deleting the highest coordinate x_{m+n-2} . The left square commutes because a vector in the kernel of $\pi_{m+n-1,m+n-2}$ has only the highest coordinate x_{m+n-2} nonzero; under $E_{m,n}$, the only surviving matrix entry is the highest-row/highest-column entry

$$\begin{pmatrix} m+n-2 \\ m-1 \end{pmatrix} x_{m+n-2},$$

which is precisely $\iota_{1,n}^m A_{m,n}(x_{m+n-2})$.

Lemma 1, with the induction hypothesis on the right vertical map, shows that $E_{m,n}$ is an isomorphic embedding. This completes the induction. $\square$

Corollary 1. *For every acceptable $\mathcal{F}$ and base point z , the space $F[3, 3]$ contains a closed subspace isomorphic to $\mathcal{F}_z^{(5)}$.*

Proof. Apply Theorem 1 with $m = n = 3$. The range $E_{3,3}[\mathcal{F}_z^{(5)}]$ is closed because $E_{3,3}$ is bounded below. $\square$

The quotient map

We now record the quotient information obtained by the same finite-order structure. This strengthens the embedding result but still does not supply a complement.

Lemma 2. *For $m \geq 2$, define*

$$\Delta_m : \mathcal{F}^{(m)} \longrightarrow \mathcal{F}^{(m-1)}, \quad \Delta_m(c_{m-1}, \dots, c_0) = (d_{m-2}, \dots, d_0),$$

where

$$d_j = c'_j - (j+1)c_{j+1} \quad (0 \leq j \leq m-2).$$

Then Δ_m is a bounded onto operator and

$$\ker \Delta_m = \{\tau_{(m,0]} f : f \in \mathcal{F}\}.$$

Proof. Use the defect-coordinate isomorphism from Lemma 6.1 of the source paper: a tuple $(c_{m-1}, \dots, c_0) \in \mathcal{F}^{(m)}$ is equivalently described by $u_0, \dots, u_{m-1} \in \mathcal{F}$, where

$$c_j = \sum_{0 \leq q \leq j} \frac{u_q^{(j-q)}}{(j-q)!} \quad (0 \leq j < m),$$

with equivalent norm $\sum_q \|u_q\|_{\mathcal{F}}$. A direct calculation gives

$$d_j = c'_j - (j+1)c_{j+1} = - \sum_{0 \leq q \leq j} \frac{(q+1)u_{q+1}^{(j-q)}}{(j-q)!}.$$

Thus, in the same defect coordinates for $\mathcal{F}^{(m-1)}$,

$$\Delta_m(u_0, u_1, \dots, u_{m-1}) = (-u_1, -2u_2, \dots, -(m-1)u_{m-1}).$$

This formula proves boundedness, surjectivity, and that the kernel consists exactly of the tuples with $u_1 = \dots = u_{m-1} = 0$, namely the tuples $\tau_{(m,0]} u_0$. $\square$

For $m, n \geq 2$, define

$$D_{m,n} : F[m, n] \longrightarrow F[m-1, n-1]$$

by

$$D_{m,n}(\tau_{(n,0]} C(z)) = \tau_{(n-1,0]} (\Delta_m C)(z).$$

This is well defined because Δ_m is first order: if C has zero n -jet at z , then $\Delta_m C$ has zero $(n-1)$ -jet at z . It is bounded by the quotient norm definition and Lemma 2.

In matrix coordinates,

$$(D_{m,n} M)_{i,j} = (i+1)M_{i+1,j} - (j+1)M_{i,j+1}, \quad 0 \leq i < n-1, \ 0 \leq j < m-1.$$

Lemma 3. *For scalar matrices, the sequence*

$$0 \longrightarrow \mathbb{C}^{m+n-1} \xrightarrow{E_{m,n}} \mathbb{C}^{n \times m} \xrightarrow{D_{m,n}} \mathbb{C}^{(n-1) \times (m-1)} \longrightarrow 0$$

is exact, where $E_{m,n}(x)_{i,j} = \binom{i+j}{i} x_{i+j}$ and $D_{m,n}$ is the formula above.

Proof. The identity $D_{m,n}E_{m,n} = 0$ is Pascal's relation in the form

$$(i+1) \binom{i+j+1}{i+1} = (j+1) \binom{i+j+1}{i}.$$

If $D_{m,n}M = 0$, the recurrence

$$(i+1)M_{i+1,j} = (j+1)M_{i,j+1}$$

forces all entries on the same anti-diagonal $i+j=s$ to have the binomial proportions

$$M_{i,j} = \binom{s}{i} x_s$$

for a uniquely determined scalar x_s . Hence $\ker D_{m,n} \subseteq \text{ran } E_{m,n}$, and the reverse inclusion has already been shown. Since

$$nm - (m+n-1) = (n-1)(m-1),$$

the rank of $D_{m,n}$ is the full dimension of the target, so $D_{m,n}$ is onto. $\square$

Theorem 2. *For $m, n \geq 2$, the map $D_{m,n}$ is a bounded quotient map and*

$$\ker D_{m,n} = E_{m,n}[\mathcal{F}_z^{(m+n-1)}].$$

Consequently,

$$F[m, n] / E_{m,n}[\mathcal{F}_z^{(m+n-1)}] \simeq F[m-1, n-1].$$

Proof. Surjectivity follows from Lemma 2: given $y \in F[m-1, n-1]$, choose $H \in \mathcal{F}^{(m-1)}$ with $\tau_{(n-1,0]}H(z) = y$. Since Δ_m is onto, choose $C \in \mathcal{F}^{(m)}$ with $\Delta_m C = H$. Then $D_{m,n}(\tau_{(n,0]}C(z)) = y$. Thus $D_{m,n}$ is a bounded onto operator, hence a quotient map.

The inclusion $E_{m,n}[\mathcal{F}_z^{(m+n-1)}] \subseteq \ker D_{m,n}$ is immediate from the formula $D_{m,n}E_{m,n} = 0$.

Conversely, let $M = \tau_{(n,0]}C(z) \in \ker D_{m,n}$. Lemma 3 gives a formal $(m+n-1)$ -jet $x = (x_{m+n-2}, \dots, x_0)$ with $M = E_{m,n}x$. It remains to see that this formal jet belongs to $\mathcal{F}_z^{(m+n-1)}$.

Write C in the defect coordinates used in Lemma 2, with $u_0, \dots, u_{m-1} \in \mathcal{F}$. The coordinates x_s obtained from the scalar recurrence are universal linear combinations of the derivatives at z of the functions u_q , with coefficients depending only on m, n . Equivalently, there are scalar polynomials $p_0, \dots, p_{m-1}$, depending only on m, n and z , such that the function

$$f = \sum_{q=0}^{m-1} p_q u_q$$

satisfies

$$\tau_{(m+n-1,0]}f(z) = x.$$

The acceptable-space module property puts $f \in \mathcal{F}$. Hence $x \in \mathcal{F}_z^{(m+n-1)}$, and $M \in E_{m,n}[\mathcal{F}_z^{(m+n-1)}]$. $\square$

Verification and limitations

The proof uses only the standard Rochberg exact sequences, the defect-coordinate description of $\mathcal{F}^{(m)}$, and the explicit matrix formula for $E_{m,n}$ from the source paper. It avoids any claim that arbitrary finite matrices on the ambient product space restrict to bounded maps between Rochberg spaces.

The result is still weaker than the source’s tempting conjecture because it does not show that $E_{m,n}$ has complemented range. The quotient is identified by Theorem 2, but no bounded linear section of $D_{m,n}$ is claimed. In particular, it should be reviewed as a positive answer to the printed $E_{3,3}$ -embedding doubt and as a quotient identification, not as a proof of the full complemented-range formula.

Novelty check

The run indexes were searched for arXiv:1906.06677, the source title, $E_{3,3}$, $F[3,3]$, $F[m,n]$, Rochberg spaces, centralizers, differentials, and complex interpolation. Existing run work on arXiv:1406.6723 concerns a different reciprocal question about trivial first differentials and nontrivial higher differentials. Web searches for exact phrases involving $E_{3,3}$, $F[3,3]$, $F[m,n]$, and Rochberg spaces found the source paper but no separate later answer. The quotient-map strengthening was added after the first embedding packet by isolating the first-order defect map.

References

- F. Cabello Sanchez, J. M. F. Castillo, and W. H. G. Correa, *Higher order derivatives of analytic families of Banach spaces*, arXiv:1906.06677.
- R. Rochberg, *Higher order estimates in complex interpolation theory*, cited in the source paper for the original higher-order spaces.
- F. Cabello Sanchez, J. M. F. Castillo, and N. J. Kalton, work cited as CCK in the source paper for the exact sequences and higher differentials.

Human-review recommendation

Review as a likely valid partial result. The key points to check are the indexing in the two commutative squares, the defect-coordinate computation in Lemma 2, and the final step in Theorem 2 realizing the formal kernel jet by polynomial multiples of the defect-coordinate functions. No complemented-range conclusion is claimed.